Jung Who Nam — Ph.D. in Computer Science

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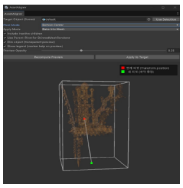
Summary

Ph.D. in Computer Science with 10+ years of experience building and developing XR applications and 3D user interfaces. Hands-on proficiency in Unity and C# to build and deploy novel spatial applications, complemented by experience with C++ and Python for systems integration. Currently researching the challenge of integrating XR technologies with desktop knowledge work. Eager to apply this R&D and prototyping experience to build the next generation of immersive productivity tools.

Relevant Experience

2024.10-Present — **Assistant professor** — **Kunsan National University – Gunsan, South Korea**

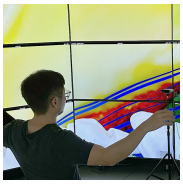


- Instruct undergraduate courses in interactive computer graphics and game engine programming.
- Conduct research on approaches to integrate XR technologies for knowledge work in a desktop setting.
- Mentor students on career development and academic progress; evaluate senior graduation projects.

Unity | C# | Java

- J. Jang and **J. W. Nam** , "A vision-based interactive system for dementia prevention hand exercises," Korean Institute of Smart Media Posters, 2025
  YouTube
- J. Jang and **J. W. Nam**, "Assetaligner: An automated 3d model normalization tool for interactive exhibitions," Korea Software Congress Posters, 2025
  YouTube
- J. Jang, S. Song, I. Kim, and **J. W. Nam**, *3D Data Visualization Utilizing Game Engines*. Hanul Media, 2025, isbn: 979-11-94649-24-3 (03560)

2022-2024 (1 yrs 10 mos) — **Postdoctoral Researcher** — **University of Texas – Austin, TX**

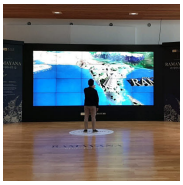


- Upgraded an open-source ray tracing application by integrating VR features, enabling immersive exploration and gesture-based navigation.

C++ | CMake | OpenMP | Docker | Intel OSPRay | Microsoft Kinect | Cluster-driven Displays | Python | Open3D

- **J. W. Nam**, G. D. Abram, F. Samsel, and P. A. Navrátil, "Immersive ospray: Enabling vr experiences with ospray," in *2023 ACM conference on practice and experience in advanced research computing (PEARC)*, 2023, p. 226–230. doi: 10.1145/3569951.3597579

2019-2021 (2 yrs 6 mos) — **Research Engineer** — **Gwangju Institute of Science and Technology – Gwangju, S. Korea**

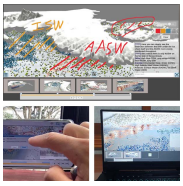


- Developed interactive museum installations in collaboration with graphic designers and curators.
- Designed gesture-based interaction techniques for exploring archived data.

Unity | C# | Microsoft Kinect | Large Format Displays

- "The Road of Ramayana." Interactive installation at Asia Culture Center, S. Korea, December 2020.
  YouTube News News
- N. Park, Y. Hong, H. Park, **J. W. Nam**, K. Kim, J. Pyo, K. Gil, and K. Lee, "Effects of age and motivation for visiting on ar museum experiences," ACM VRST Posters, 2019. doi: 10.1145/3359996.3364711

Summer 2018 (3 mos) — **Research Intern** — **INRIA – Saclay, France**



- Developed frameworks for data storytelling and collaboration across different devices (web, mobile, PC).

Unity | C# | PHP | MySQL

- **J. W. Nam**, T. Isenberg, and D. F. Keefe, "V-mail: 3d-enabled correspondence about spatial data on (almost) all your devices," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 4, pp. 1853–1867, 2024. doi: 10.1109/TVCG.2022.3229017
  YouTube

2014-2019
(4 yrs 9 mos)

Research Assistant — University of Minnesota – Minneapolis, MN

- Designed and developed 3D interactive tools to assist scientists with data analysis and presentation.
- Collaborated with teams from the U.S. National Forest Services and the Medical Device Center.

Unity | C# | OptiTrack | MS Kinect | HMD VR | CAVE VR | Mobile VR | Projection-based AR | Processing | 3D Printing

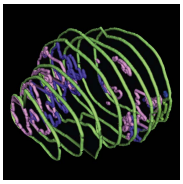
- **J. W. Nam** et al., "Worlds-in-wedges: Combining worlds-in-miniature and portals to support comparative immersive visualization of forestry data," in *2019 IEEE conference on virtual reality and 3D user interfaces (VR)*, 2019, pp. 747–755. doi: 10.1109/VR.2019.8797871
  YouTube Presentation IEEE VR
- **J. W. Nam**, C. H. Perry, B. T. Wilson, and D. F. Keefe, "Linked view visualization using clipboard-style mobile vr: Application to communicating forestry data," IEEE VIS Posters, 2019
  YouTube SciVis Best Poster Award
- **J. W. Nam** and D. F. Keefe, "Spatial correlation: An interactive display of virtual gesture sculpture," *Leonardo*, vol. 50, no. 1, pp. 94–95, 2017. doi: 10.1162/LEON_a_01226
  YouTube

2011-2014
(3 yrs 3 mos)

Programmer — University of Minnesota – Minneapolis, MN

- Developed a Java-based image processing application for assembling scanned pathology images and annotating cancer boundaries.

Java | Java3D



- E. L. et al., "Signature maps for automatic identification of prostate cancer from colorimetric analysis of h&e-and ihc-stained histopathological specimens," *Nature Scientific Reports*, vol. 9, no. 6992, 2019. doi: 10.1038/s41598-019-43486-y
- G. J. M. et al., "Detection of prostate cancer: quantitative multiparametric mr imaging models developed using registered correlative histopathology," *Radiology*, vol. 279, no. 3, pp. 805–816, 2016. doi: 10.1148/radiol.2015151089

Professional Services

Panelist The Future of AI that Communicates with Humans, Hwangryong Humanities Talk Concert, 2025.10.31

Reviewer ISMAR(2024), TVCG (2024, 2023), VR (2022), DESRIST (2018), Leonardo (2016)

Education

2014-2022 — **Ph.D., Computer Science** — University of Minnesota – Minneapolis, MN

- Advisor: Prof. Daniel F. Keefe
- Committee: Dr. Tobias Isenberg, Prof. Lana Yarosh, Prof. Evan Suma Rosenberg
- Dissertation title: *Everyday Scientific Visualization: Making 3D Visualization Techniques Accessible for Day-To-Day Team-Science for Collaboration and Analysis*
- Specializations: Data visualization, virtual reality

2012-2018 — **M.S., Computer Science** — University of Minnesota – Minneapolis, MN

- Specializations: Computer graphics, virtual reality

2008-2012 — **B.S., Computer Science** — University of Minnesota – Minneapolis, MN

- Specializations: Computer graphics, user interfaces

Technical Proficiencies

Programming Languages	Advanced - C#	Familiar - C++, Python, Java
Development Tools	Advanced - Unity	Familiar - Intel OSPRay, Open3D, CMake, OpenMP, Docker
3D Tracking Systems	OptiTrack, Microsoft Kinect, Leap Motion	
Display Technologies	HMD VR, CAVE VR, Projection-based AR, Cluster-driven Wall Displays	